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발표 자료

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김한서

이번 주 진행사항

- FACT: Fine-Grained Across-Variable Convolution for Multivariate Time-Series Forecasting
 - 논문 리뷰

FACT: FINE-GRAINED ACROSS-VARIABLE CONVOLUTION FOR MULTIVARIATE TIME-SERIES FORECASTING

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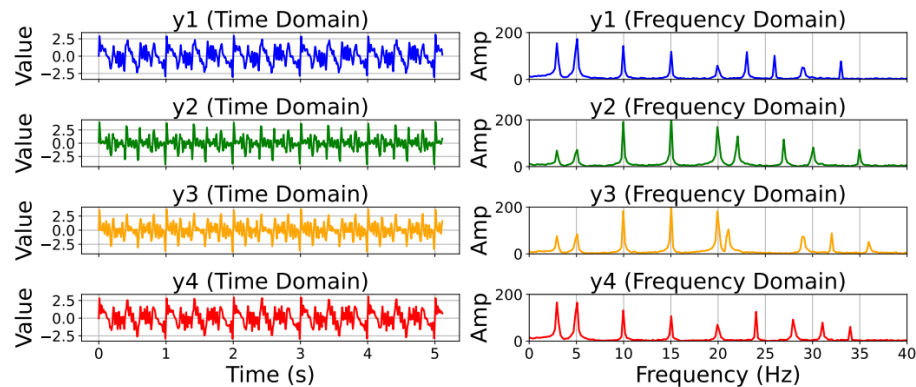
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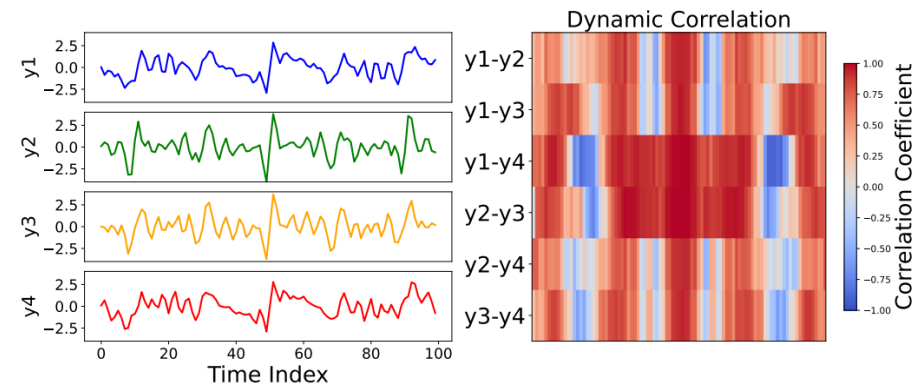
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- Published at ICLR 2026
- 링크: https://openreview.net/forum?id=j3gNYqrHtl&utm_source=chatgpt.com

Background

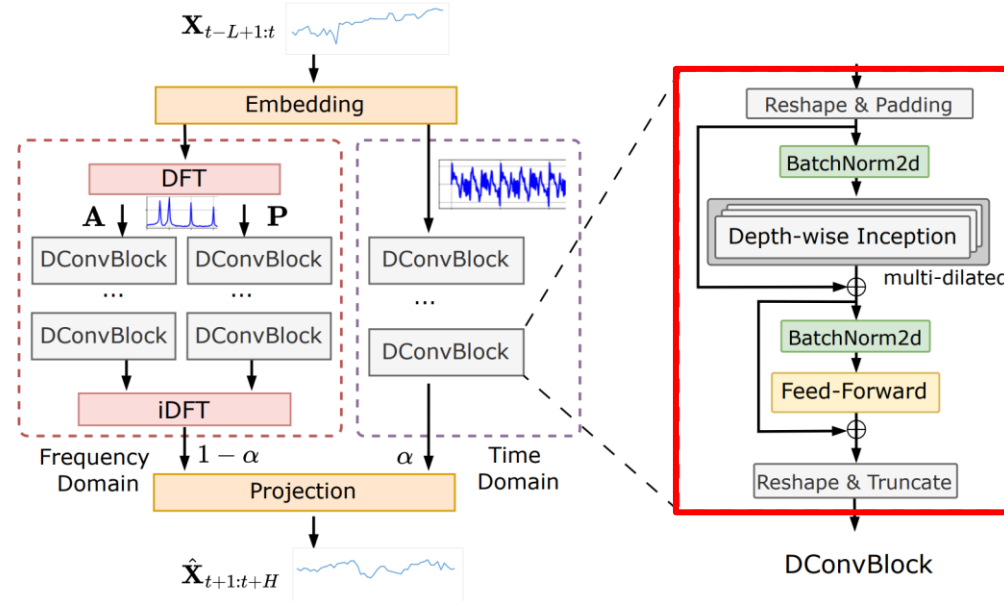


(a) Each frequency.

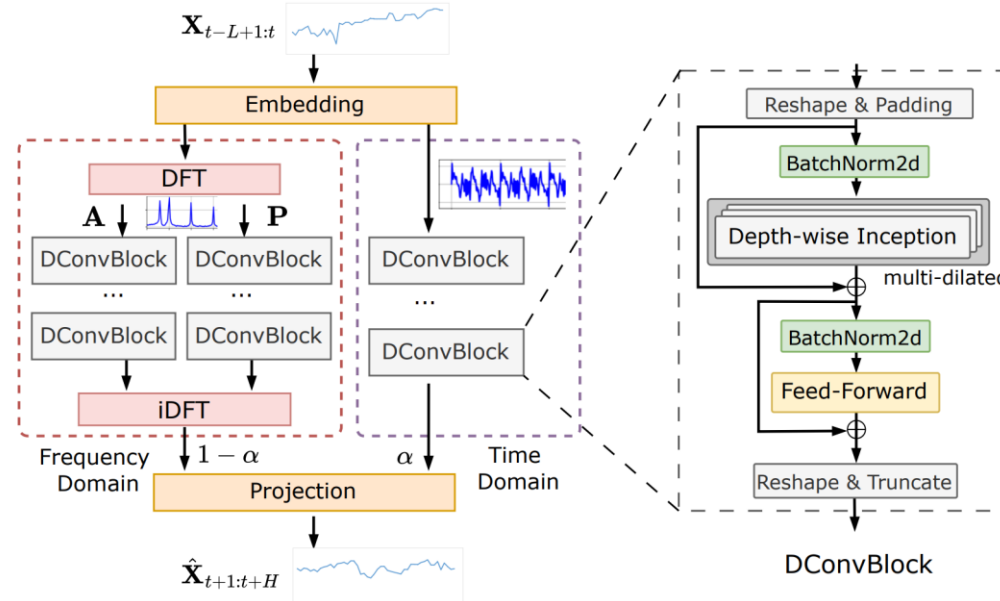


(b) Each time step.

- 기존 다변량 시계열 예측은 변수 간 관계를 coarse-grained하게 포착, 시간 및 주파수에 따라 동적으로 변화하는 세밀한 변수 간 상호작용을 충분히 반영하지 못함
- FACT 논문은 DConvBlock을 통해 시간에 따라 변화하는 변수 관계와 주파수별 변수 관계를 각각 fine-grained하게 모델링



- **Embedding**
 - 다변량 시계열 데이터를 D 차원 임베딩 벡터로 변환 $X_{t-L+1:t} \in \mathbb{R}^{C \times L} \rightarrow X_e \in \mathbb{R}^{C \times D}$
- **DConvBlock**
 - Reshape를 통해 2D로 변경 후 Batch normalization 진행, multi-dilated Depth-wise Inception으로 다양한 범위의 변수 간 상호작용을 포착
 - Batch normalization 및 FFN을 거친 뒤, 원래 차원으로 복원



• Frequency Domain

- 주파수 성분의 진동 세기를 나타내는 amplitude A 와 상대적인 timing을 나타내는 phase P 를 각각 DConvBlock을 통해 주파수별 세밀한 변수 간 상호작용을 모델링

$$X_e \xrightarrow{\text{DFT}} F = R + jI, A = |F|, P = \angle F$$

• Time Domain

- 임베딩 벡터에 DConvBlock을 적용하여 시간 도메인에서 세밀한 변수 간의 동적 상호작용을 모델링

주요 모델 성능 비교

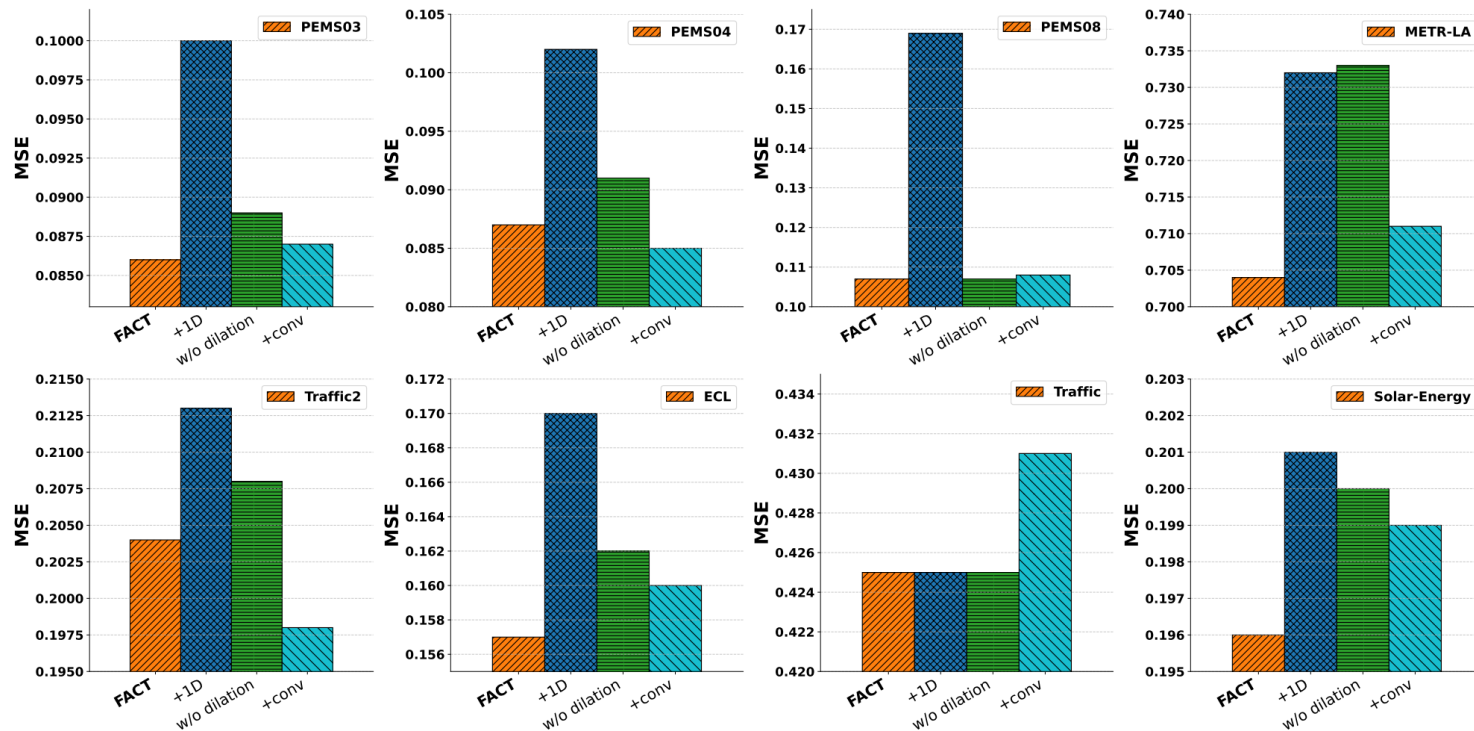
Models	ECL		Traffic		Solar-Energy		electricity		Traffic2		METR-LA		1 st
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
TimesNet(2023a)	0.192	0.295	0.620	0.336	0.301	0.319	965.69	2.265	0.272	0.258	0.908	0.520	0
PatchTST(2023)	0.189	0.276	0.454	0.286	0.236	0.266	410.90	1.648	0.234	0.204	0.762	0.488	0
CycleNet(2024a)	0.168	0.259	0.472	0.301	0.210	0.261	178.53	0.988	0.292	0.254	0.748	0.503	0
SOFTS(2024)	0.174	0.264	0.409	0.267	0.229	0.256	264.81	1.304	0.224	0.192	0.748	0.503	1
TSMixer(2025)	0.186	0.287	0.522	0.357	0.260	0.297	257.08	1.222	0.244	0.232	0.833	0.490	0
ModernTCN(2024)	0.197	0.282	0.546	0.348	0.211	0.310	509.11	1.899	0.252	0.245	0.808	0.500	0
TimePro(2025)	0.169	0.262	0.443	0.287	0.232	0.266	200.68	1.093	0.244	0.218	0.787	0.533	0
DUET(2025)	0.170	0.265	0.469	0.289	0.220	0.263	244.13	1.244	0.279	0.244	0.813	0.487	0
TQNet(2025)	0.164	0.259	0.445	0.276	0.198	0.256	203.41	1.038	0.234	0.200	0.749	0.489	0
iTransformer(2024)	0.178	0.270	0.428	0.282	0.233	0.262	384.22	1.656	0.221	0.193	0.837	0.536	0
TimeXer(2024b)	0.171	0.270	0.466	0.287	0.202	0.269	261.84	1.310	0.217	0.195	0.750	0.496	0
TimeFilter(2025)	0.158	0.256	0.407	0.268	0.223	0.250	193.29	1.054	0.227	0.199	0.819	0.493	2
FACT	0.157	0.254	0.425	0.276	0.196	0.254	159.73	0.922	0.204	0.179	0.704	0.456	9

- FACT 모델 성능이 기존 모델들보다 더 좋은 모습을 보임

Models	PEMS03		PEMS04		METR-LA		ECL		Traffic		Solar		Traffic2	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
FACT	0.086	0.194	0.087	0.194	0.704	0.456	0.157	0.254	0.425	0.276	0.197	0.259	0.204	0.179
w/o Time	0.089	0.197	0.089	0.198	0.755	0.483	0.162	0.258	0.436	0.286	0.202	0.263	0.215	0.188
w/o Frequency	0.087	0.196	0.090	0.201	0.722	0.459	0.162	0.258	0.426	0.276	0.198	0.256	0.212	0.180

- 시간 및 주파수 도메인을 모두 포함했을 때의 성능이 가장 좋은 것을 확인

Ablation Study



- FACT 모델이 전반적으로 좋은 성능을 보였으며, 2D가 아닌 1D Conv일 때의 성능 하락이 가장 큼

부록

주요 모델 성능 비교

Models	PEMS03		PEMS04		PEMS07		PEMS08		PEMSD7		PEMS-BAY		1 st
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	
TimesNet(2023a)	0.173	0.264	0.148	0.263	0.133	0.241	0.214	0.293	0.569	0.436	0.669	0.375	0
PatchTST(2023)	0.150	0.260	0.171	0.283	0.165	0.267	0.242	0.285	0.485	0.412	0.596	0.382	0
CycleNet(2024a)	0.118	0.225	0.119	0.231	0.113	0.189	0.150	0.228	0.503	0.429	0.604	0.397	0
SOFTS(2024)	0.104	0.210	0.102	0.208	0.087	0.184	0.138	0.219	0.503	0.429	0.604	0.397	0
TSMixer(2023)	0.158	0.263	0.129	0.247	0.117	0.221	0.201	0.294	0.488	0.394	0.523	0.336	0
ModernTCN(2024)	0.111	0.226	0.103	0.217	0.118	0.208	0.230	0.268	0.374	0.370	0.436	0.330	0
TimePro(2025)	0.119	0.225	0.114	0.226	0.096	0.200	0.252	0.259	0.412	0.370	0.577	0.389	0
DUET(2025)	0.110	0.219	0.113	0.227	0.091	0.198	0.130	0.229	0.360	0.336	0.447	0.326	0
TQNet(2025)	0.097	0.203	0.091	0.197	0.075	0.171	0.142	0.229	0.361	0.343	0.427	0.315	0
iTransformer(2024)	0.113	0.221	0.119	0.231	0.113	0.189	0.150	0.228	0.382	0.354	0.447	0.319	0
TimeXer(2024b)	0.111	0.223	0.110	0.227	0.093	0.199	0.233	0.269	0.337	0.319	0.394	0.288	0
TimeFilter(2025)	0.108	0.217	0.114	0.226	0.101	0.207	0.139	0.230	0.498	0.404	0.547	0.358	0
FACT	0.086	0.194	0.087	0.194	0.073	0.170	0.107	0.200	0.330	0.316	0.372	0.267	12

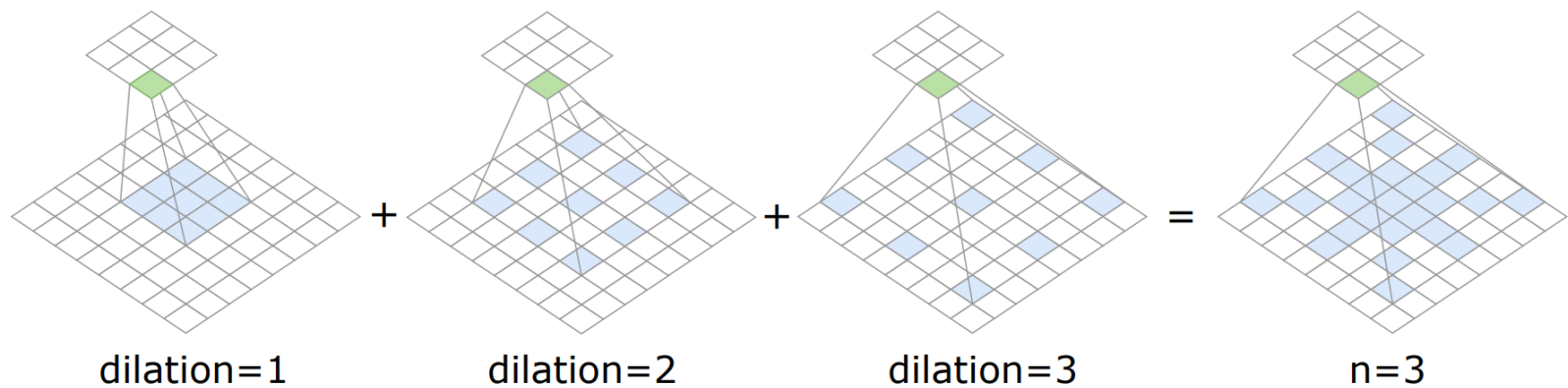


Figure 3: Multi-dilated architecture. $n = 3$ denotes 3 increasing dilation sizes.